When Does Adaptive Recovery Pay? A Failure-Aware Bandit Study of Recovery-Strategy Selection in Autonomous LLM Agents

Pio Micheal
Department of Artificial Intelligence and Data Science
Karunya Institute of Technology and Sciences
Coimbatore, India
Email: piomicheal17@gmail.com

***Abstract*—Autonomous agents that call external tools fail in recurring, classifiable ways, and a growing body of work treats the choice of recovery strategy as a decision problem rather than a fixed exception handler. If that choice is learned online, a practical question follows: at what horizon does a learner overtake a competent hand-written rule table? An existing predictor in our framework answers this question under the assumption that the policy observes the true failure type. That assumption does not hold in our implementation, where every policy sees only a classifier's predicted type. We extend the predictor to the noisy-classifier setting by re-expressing the strategy-success matrix in predicted-type space through an exact Bayes mixture, and we separate the resulting degradation into a reward-ceiling component and a learnability component. At the measured accuracy of our trained classifier (0.939 mean diagonal), ceiling loss dominates learnability loss by roughly a factor of six. We also show, by construction, that the predicted crossover horizon is not monotonic in classifier accuracy in general, even though the static-side regret rate is. The extension's free multiplicative constant was evaluated under a protocol registered before the supporting data existed: an environment-level 18/8 train/test split, a transfer branch and an in-condition branch fitted on the same training environments, and a pre-committed decision threshold. The observed ratio of held-out errors was $R = 1.4729$ against a threshold of 1.5, so the pre-committed outcome is the favourable one, but the same protocol's bootstrap confidence interval for R is $[1.00, 9.55]$ and does not confirm that the threshold is cleared robustly. We therefore report the component as partially supported and describe it as a calibrated predictor rather than a validated bound. A companion investigation into whether a low-dimensional statistic can predict the sign of adaptive advantage under environment drift is reported as a negative result: three pre-registered candidates, including one retracted for conceptual circularity, produced no sign rule that survives leave-one-cell-out evaluation, though one qualitative finding does survive: the sign is governed by whether staleness damage reverts, not by its magnitude.**

***Index Terms*—autonomous agents, failure recovery, multi-armed bandits, Thompson sampling, regret analysis, classifier noise, pre-registration, negative results, reproducibility**

I. Introduction

Tool-using language agents fail often enough that failure handling is a first-class design concern rather than an edge case. Timeouts, tool exceptions, malformed outputs, and lost context are not rare pathologies; recent empirical work documents them as the dominant modes in deployed and benchmarked systems alike [7], [8], [9]. The usual engineering response is a fixed mapping: when failure type t occurs, apply recovery strategy $\pi(t)$. Such a table is cheap, interpretable, and often good. Its weakness is that it never improves, and it is only as good as the designer's beliefs at the moment it was written.

The alternative is to treat each (failure type, recovery strategy) pair as an arm whose value is estimated from outcomes, and to select strategies with a bandit algorithm. This buys adaptation at the cost of exploration: early on, the learner is worse than a good table, because it is still paying to find out what works. Whether adaptation is worth buying therefore depends on the deployment horizon, and a designer choosing between the two approaches benefits from an estimate of the crossover point.

An earlier component of this project, which we refer to throughout as the base predictor, supplies such an estimate. It computes a static table's constant per-episode regret rate exactly, models the learner's cumulative regret with an asymptotic bound in the tradition of Lai and Robbins [1], and solves for the horizon at which the two curves meet. That construction assumes the policy acts on the true failure type.

Our implementation does not satisfy that assumption. Tracing the call chain from the episode loop through the adapter and selector layers to the matrix environment confirms that no policy in the repository has access to a true type: the context object carries no field from which one could be read. Every policy acts on a classifier's prediction. This is a structural fact about the simulator, not a modelling choice, and it means the base predictor was answering a question about a system we do not have.

This paper reports two extensions of the base predictor and their outcomes. The contributions are:

1) An exact reformulation of the crossover problem under an arbitrary classifier confusion matrix, obtained by mapping the success-probability matrix into predicted-type space by Bayes' rule. The reformulation reduces exactly to the base predictor when the classifier is perfect, and we separate which of its components are

exact and which inherit the base predictor's asymptotic approximation.
2) A decomposition of classifier-induced degradation into two channels (reward-ceiling loss and gap shrinkage), measured at our trained classifier's empirical confusion matrix, together with a construction showing that the predicted crossover horizon is not monotonic in classifier accuracy in general.
3) A pre-registered, environment-level held-out evaluation of the extension's free constant, executed against a 26-environment dataset generated after the protocol was committed. We report the outcome, including the uncertainty quantification that the protocol itself required and that does not support a strong reading of the result.
4) A documented negative result on whether the sign of adaptive advantage under environment drift reduces to a low-dimensional summary statistic, including a retracted intermediate attempt and one qualitative finding that survives.
5) A set of measurement findings about cost estimation for long parallel sweeps, which invalidate the linear-speedup assumption underlying our own published cost figures.

We are deliberate about what is not claimed. The extension does not make the underlying asymptotic bound accurate; held-out absolute errors remain large (roughly 40–60%). What the evaluation tests is whether the classifier-noise reformulation introduces *additional* model error beyond what the base bound already carries, and even that conclusion rests on a point estimate whose interval is wide.

II. Background and Related Work

A. Failure taxonomies and recovery in agent systems

Recent work has moved from anecdotal failure reports toward structured taxonomies. Cemri et al. derive an empirically grounded taxonomy of multi-agent failure modes from annotated execution traces, organised into categories spanning system design, inter-agent misalignment, and task verification [7]. Zhu et al. take a single-agent, module-oriented view, attributing failures to memory, reflection, planning, action, and system-level operations, and pair the taxonomy with a debugging framework that isolates the earliest error whose correction changes the outcome [8]. Song et al. study the complementary direction of optimising the environment rather than the agent, and likewise propose a taxonomy of agent-environment interaction failures [10].

Self-correction methods address the response to failure. Reflexion has agents verbalise reflections on failed attempts and retain them in an episodic buffer to inform later trials [5]. CRITIC has the model consult external tools to verify and revise its own outputs [6]. PALADIN trains recovery behaviour directly, using systematically injected tool failures and recovery-annotated trajectories [9].

The framework this paper extends sits at a different point. It does not propose a new self-correction mechanism; it asks how a recovery-strategy *selector* should be evaluated against a static baseline, and treats the selection itself as a bandit problem with persistent outcome memory. The present paper is narrower still: it is about the analytic prediction of when that selector overtakes the baseline, and about what happens to that prediction when the failure type fed to the selector is itself noisy.

TABLE I
Failure taxonomy ($N = 7$) and recovery strategy bank ($K = 6$).

Failure type	Meaning
`tool_timeout`	Tool call does not return in time
`tool_error`	Tool returns an error or exception
`wrong_tool_selected`	Agent chooses an inappropriate tool
`hallucinated_output`	Agent fabricates unsupported information
`malformed_output`	Output has invalid or incomplete structure
`context_loss`	Agent loses earlier instructions or state
`ambiguous_instruction`	Instruction admits several readings
Strategies: `retry`, `replan`, `switch_tool`, `re_ground`, `verify`, `clarify`	

B. Bandit background

The exploration-exploitation trade-off in the stochastic multi-armed bandit is classical. Lai and Robbins established that regret must grow at least logarithmically in the number of plays and gave asymptotically efficient allocation rules [1]; the regret model used by our base predictor descends from this result and inherits its asymptotic character. Auer et al. later showed that logarithmic regret is achievable uniformly in time with simple index policies, of which UCB1 is the canonical instance [2]. Thompson sampling, the posterior-sampling heuristic originally proposed for comparing two unknown probabilities [3], is the method whose crossover behaviour we analyse here; Russo et al. give a modern treatment [4].

The relevant consequence for this paper is that the asymptotic bound carries a multiplicative constant that cannot be derived from first principles for a given problem instance. In the base predictor this constant is denoted c; in the extension we call it κ. Everything downstream of it is an empirical calibration exercise, not a proof.

III. System Context

We summarise only the parts of the surrounding framework that the analysis depends on. The framework injects failures into simulated agent episodes, classifies each failure, selects a recovery strategy with a bandit algorithm (Thompson sampling, UCB1, or epsilon-greedy), records outcomes in persistent memory, and evaluates against static baselines.

A. Taxonomy and strategy bank

Failures are drawn from a seven-type taxonomy and recovered with one of six strategies, giving $N = 7$ and $K = 6$ throughout this paper (Table I).

B. Canonical environment and static baseline

A canonical environment fixes a 7×6 matrix P, where $P[t, k]$ is the success probability of strategy k for true failure

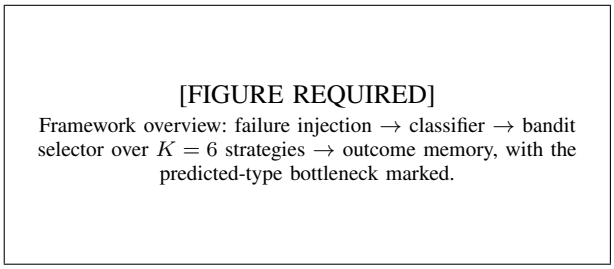


Fig. 1. Position of the classifier in the decision path. Policies observe only the predicted type t', never the true type t; this is the structural fact that motivates the extension in Section V.

type t, together with a hand-authored static mapping. That mapping agrees with the row-wise argmax of P on six of the seven types, which we describe as quality $6/7$: it is a deliberately competent baseline, not a straw man. All results below that are not explicitly swept or sampled use this pair.

C. The base crossover predictor

The base predictor supplies three objects. First, $\rho(P, \pi, p)$, a static table's constant per-episode regret rate under prior p over failure types, computed exactly as the difference between the oracle rate and the policy rate. Because a static table pays this penalty every episode indefinitely, its cumulative regret is the line ρT. Second, $R_{\mathrm{TS}}(T)$, an asymptotic upper bound on Thompson sampling's cumulative regret, available in a KL-divergence form and a cheaper gap form, and carrying the free constant c. Third, the crossover horizon T^*, the smallest T solving $R_{\mathrm{TS}}(T) = \rho T$, found by scanning for the first sign change of $h(T) = R_{\mathrm{TS}}(T) - \rho T$ and refining with a bracketing root finder; h is unimodal, so the scan cannot return a spurious second root.

Calibration of the base predictor across 56 empirical cells produced a best fit of $c^* \approx 0.00452$, roughly a $220\times$ downward correction from the literal $c = 1.0$. Even at that value the predictor was found to be reliable only in a narrow regime, namely poor tables (quality $\leq 1/7$) at small gaps (around 0.05). At the canonical table's own quality of $6/7$, the calibrated predictor underestimates the observed crossover by roughly $50\times$. This is the single most important inherited caveat: the predictor we extend was not fully validated even in its simpler, noise-free form, and no result in this paper repairs that.

IV. Problem Statement

Let $\mathrm{confusion}[t, t'] = \Pr(\text{predicted } t' \mid \text{true } t)$ describe the classifier. The base predictor implicitly assumes $\mathrm{confusion} = I$. Our system does not. Two questions follow.

First, what is the correct crossover prediction when the policy conditions on t' rather than t? This is a modelling question with an exact answer for the static side and an approximate one for the adaptive side.

Second, does the free constant fitted under one classifier condition transfer to another? If a single κ fitted on perfect-classifier data predicts noisy-classifier held-out cells about as well as a κ fitted directly on noisy-classifier data, then the reformulation has absorbed the classifier effect and the residual error is inherited. If not, the reformulation introduces error of its own. Distinguishing these two possibilities requires genuinely held-out data and a decision rule fixed in advance, for reasons we return to in Section VI.

V. Proposed Methodology

A. The effective mixture matrix

Marginalising over true types gives the predicted-type distribution

$$p'[t'] = \sum_t p[t] \cdot \mathrm{confusion}[t, t']. \tag{1}$$

Applying Bayes' rule inside each predicted bucket gives the effective success probability of strategy k given that the classifier predicted t':

$$P_{\mathrm{eff}}[t', k] = \frac{\sum_t p[t]\,\mathrm{confusion}[t, t']\,P[t, k]}{p'[t']}. \tag{2}$$

Equations (1) and (2) are exact; no approximation enters. When $\mathrm{confusion} = I$ they reduce to $p' = p$ and $P_{\mathrm{eff}} = P$ identically. P_{eff} is the central object of the extension: it answers "given that the classifier said t', how well does strategy k actually work, averaged over which type it really was."

B. Static and adaptive sides under confusion

The static side follows by composing two existing exact functions:

$$\rho_{\mathrm{conf}} = \mathrm{oracle}(P, p) - \mathrm{policy}_{\mathrm{conf}}(P, \pi, \mathrm{confusion}, p), \tag{3}$$

which is exact, not asymptotic. The adaptive side reuses the base bound unmodified, evaluated on the mixture instead of the true matrix:

$$R_{\mathrm{TS}}^{\mathrm{conf}}(T) = R_{\mathrm{TS}}(P_{\mathrm{eff}}, T, N, p', \mathrm{form}, \kappa)\,. \tag{4}$$

The crossover horizon under confusion solves

$$R_{\mathrm{TS}}^{\mathrm{conf}}(T) = \rho_{\mathrm{conf}} \cdot T. \tag{5}$$

Equation (5) is implemented separately rather than as a wrapper around the base solver. The reason is that it mixes index spaces: ρ_{conf} lives in true-type space while the regret side lives in predicted-type space, and the base solver computes its own ρ from the same triple it uses for the regret side, so it cannot represent this combination internally. The numerical strategy (unimodal scan plus bracketing refinement) is shared. Unit tests confirm that at $\mathrm{confusion} = I$ the extension returns bit-for-bit what the base solver returns, so it is a strict generalisation rather than a divergent reimplementation.

We also report the ceiling reduction,

$$\Delta_{\mathrm{ceil}} = \mathrm{oracle}(P, p) - \mathrm{oracle}(P_{\mathrm{eff}}, p'), \tag{6}$$

which is exact, provably non-negative, and exactly zero at $\mathrm{confusion} = I$.

TABLE II
EXACT COMPONENTS VERSUS COMPONENTS INHERITING THE BASE PREDICTOR'S ASYMPTOTIC APPROXIMATION.

Component	Nature
Effective matrix, Eq. (1)–(2)	Exact
ρ_{conf}, Eq. (3)	Exact
Ceiling reduction, Eq. (6)	Exact
Gap-shrinkage report	Exact (a report)
Argmax-inversion check	Exact (a check)
$R_{\mathrm{TS}}^{\mathrm{conf}}$, Eq. (4)	Inherited asymptotic
Crossover horizon, Eq. (5)	Inherited asymptotic

C. Assumptions

The extension rests on five assumptions, listed here with their status rather than presented as uniformly benign.

A1 (predicted-type bucket structure). Verified at the implementation level by tracing the call chain, not assumed from documentation. In the usual sense it is a structural fact about the simulator rather than an assumption.

A2 (conditionally Bernoulli rewards). Inherited unchanged from the base predictor.

A3 (stationary confusion matrix). No classifier drift within the analysed horizon. This is optimistic if the classifier itself degrades over time, which we do not model.

A4 (asymptotic regret bound). Inherited unchanged. Valid as $T \to \infty$; not a finite-time guarantee; carries the uncalibrated κ.

A5 (scalar accuracy as a convenience). A uniform-confusion construction is available when a full matrix is not, and this is a real simplification. We therefore validate primarily against the trained classifier's actual, non-uniform confusion matrix rather than relying on the scalar shortcut.

D. What is exact and what is inherited

We consider this separation the more useful framing of the contribution than "a new bound." Table II states it explicitly. The extension builds exact machinery around an existing asymptotic bound and is careful about which parts are which.

E. Two channels of degradation

Classifier noise hurts in two distinguishable ways. Merging true types into indistinguishable predicted buckets destroys information and lowers the achievable reward ceiling regardless of policy; this is Eq. (6). Separately, mixing changes the gap between the best and second-best action within each predicted bucket, which affects how hard the arm-identification problem is. The first is a ceiling effect, the second a learnability effect, and they need not move together.

F. Argmax inversion

A conceivable pathology is that a predicted bucket's own best action under P_{eff} is optimal for none of the true types feeding that bucket: the mixture would then recommend something no constituent type wanted. We check for this explicitly. It does not occur at our classifier's measured confusion (Section VII), and a constructed three-type counter-example confirms the check fires when its structural condition (two comparably weighted contributors with different argmaxes and no dominant contributor) is met, so the check is not a dead code path.

G. Non-monotonicity in accuracy

ρ and the ceiling reduction are provably monotonic in classifier accuracy. T^* is not. A clean two-type, two-strategy construction, kept as a unit test, exhibits a predicted bucket whose best arm changes identity as accuracy falls through a threshold, making that bucket's top-two gap V-shaped in accuracy: large at both extremes and exactly zero at the flip. This is a property of the shape of Eq. (5) and holds independently of what κ turns out to be. It matters practically because it forbids the intuitive inference "a somewhat better classifier implies a somewhat earlier crossover."

VI. EXPERIMENTAL SETUP

A. Why a held-out protocol was necessary

An earlier version of this validation fitted κ and evaluated it on the same three environments, because only three had been simulated at the time. Those numbers are uninterpretable as validation in principle: with the fit and the evaluation drawn from one three-cell subset, a low error cannot be distinguished from tuning to those three points. The earlier analysis is retained in version control for transparency and is no longer reported as a result. Fixing it required two things: enough environments for a real split, and a protocol committed before the new data existed.

B. Closing the data gap

A read-only forensic check established that the committed sweep table had only ever contained 3 of the intended 26 environments. All 617 files in the sweep result store were loaded, not sampled, and the remaining 23 environments had no trace anywhere: a genuine never-simulated gap rather than a lost cache state or a filtering bug. The gap traced to a stale status note that had propagated into a false belief about coverage; the actual documented scope was a deliberate cost-gated reduction taken after the full grid was estimated at around 73 hours.

Closing the gap required new simulation. We filtered to the two methods the primary analysis needs (the static baseline and Thompson sampling), dropping the other nine default methods as out of scope. The filtered grid is 312 cells (26 environments $\times$ 2 classifier conditions $\times$ 2 methods $\times$ 3 seeds), of which 204 were new. The resulting dataset comprises 26 environments $\times$ 2 classifier conditions $=$ 52 cells and 2,184 rows across 7 horizons.

C. Pre-registered protocol

The protocol was written and committed before the 26-environment data existed. Its elements were as follows.

Split. Of the 25 gap-family environments, 8 were assigned to TEST by a fixed-seed draw (split seed 19001); the single non-gap-family canonical environment was placed in TRAIN

unconditionally, on the structural grounds that holding out the only member of its family would test a different question. This yields 18 TRAIN and 8 TEST environments. The protocol disclosed in advance that the two environments whose poor errors were already known from the superseded analysis had landed in TEST, and did not redraw the seed on that basis, since redrawing after observing the draw is precisely the behaviour pre-registration exists to prevent.

Fitting. Identical to the base predictor's own procedure: bounded one-dimensional least squares minimising mean squared log-space error, with bounds $(\log 10^{-3}, \log 10^{4})$ and the KL form fixed rather than varied.

Branches. Branch A (transfer) fits κ_A on TRAIN environments' perfect-classifier cells and applies it unchanged to TEST environments' noisy-classifier cells. Branch B (in-condition) fits κ_B on the same TRAIN environments' noisy-classifier cells and applies it to the same TEST cells. Branch B is the best achievable same-condition fit and serves as the benchmark.

Decision statistic and rule. $R = E_A/E_B$, where E denotes median absolute relative error on held-out predictions. $R \leq 1.5$ was pre-committed to Case 1 (the reformulation absorbs the classifier effect; residual error is inherited); $R > 1.5$ to Case 2 (the reformulation introduces its own error, shipped as a documented non-result with no rescue attempt). The threshold of 1.5 was chosen as a round middle point between a too-lenient factor of three and a too-strict factor of one, and was disclosed as a judgment call rather than derived.

Power floor. Fewer than 5 finite cells in either branch would yield no Case 1/Case 2 conclusion at all, reported instead as underpowered.

Uncertainty. Environment-level bootstrap (seed 19002, 2,000 resamples), because cells sharing an environment are not independent. Two bootstraps were specified: held-out error uncertainty, resampling TEST environments and reusing the fitted constants; and fit uncertainty, resampling TRAIN environments and refitting per resample for each branch separately.

Naming rule. If the operative fitted constant fell within one order of magnitude of the base predictor's own best fit, that is within $[0.00045, 0.045]$, the result would be described as a calibrated predictor rather than a validated bound. This was decided before the fitted value was seen.

Commitment. One specification only. No second split, form, threshold, or fitting bound would be tried after seeing results, whatever the outcome.

D. Hardware and software

Simulations ran on a machine with 4 physical and 8 logical cores, using a process-based parallel backend with a content-addressed result store that makes interrupted runs resumable. Further environment details are [DETAIL REQUIRED].

VII. Results and Discussion

A. Two-channel decomposition at the real classifier

At the trained classifier's empirical confusion matrix (mean diagonal accuracy 0.939), the ceiling reduction is 0.0120 and the mean gap shrinkage across the seven predicted buckets is 0.0020. Ceiling loss is therefore about six times larger than learnability loss at this accuracy. Both are present and neither is zero, but the interpretation is that most of the measured degradation is information destroyed by merging types rather than a harder arm-identification problem. This is specific to this classifier's confusion structure and should not be read as a general ratio.

The argmax-inversion check does not fire in any of the seven predicted buckets at this confusion matrix.

TABLE III
STATIC-SIDE VALIDATION. PREDICTED VERSUS EMPIRICAL PER-EPISODE RATE ACROSS NOISE LEVELS β; 50 SEEDS PER LEVEL.

β	Predicted	Empirical	Abs. error
0.0	0.4408	0.4306	0.0102
0.1	0.4691	0.4582	0.0109
0.3	0.5256	0.5112	0.0144
0.5	0.5821	0.5744	0.0077
0.8	0.6669	0.6584	0.0085
1.0	0.7234	0.7156	0.0078
Median absolute error			**0.0093**

TABLE IV
PRE-REGISTERED HELD-OUT EVALUATION. ERRORS ARE MEDIAN ABSOLUTE RELATIVE ERROR ON THE EIGHT HELD-OUT TEST CELLS.

	Branch A (transfer)	**Branch B** (in-condition)
Fitted constant	0.044381	0.042559
TRAIN cells used	18	18
Held-out error	59.7%	40.5%
Fit 95% CI	$[0.0369, 0.0544]$	$[0.0105, 0.0426]$
$R = E_A/E_B = 0.5970/0.4053 = \mathbf{1.4729}$		
Threshold 1.5; margin 1.8%		
Bootstrap 95% CI for R: $[\mathbf{1.00},\ \mathbf{9.55}]$		

B. Static-side validation

The exact static component of Eq. (3) was checked against an existing table of six noise levels with 50 already-simulated seeds each. Because that table contains no adaptive-method trajectory, it validates the static side only and is never used to claim the adaptive theory is validated. Results appear in Table III.

A median absolute error of 0.93 percentage points is the strongest validation result anywhere in this work, which is what one would expect of a component that is exact closed-form arithmetic rather than an asymptotic approximation. It says nothing about the adaptive side.

C. Held-out evaluation of the free constant

The power check passed with margin: 18 of 18 TRAIN cells and 8 of 8 TEST cells finite in both branches, against a floor of 5. Headline results are in Table IV.

The point estimate satisfies the pre-committed rule, so the reported outcome is Case 1, and the pre-committed Case 1 wording applies: the mixture reformulation captures

TABLE V
BRANCH A PER-CELL RESULTS ON THE EIGHT HELD-OUT TEST ENVIRONMENTS.

Environment	Ratio	Signed rel. error
gap0.20_idx00	1.487	$+48.7\%$
gap0.20_idx01	1.527	$+52.7\%$
gap0.20_idx06	1.283	$+28.3\%$
gap0.20_idx08	5.756	$+475.6\%$
gap0.20_idx11	0.675	-32.5%
gap0.20_idx16	0.333	-66.7%
gap0.20_idx17	5.752	$+475.2\%$
gap0.20_idx24	1.694	$+69.4\%$

TABLE VI
COMPONENT CLASSIFICATION.

Component	**Status**
Effective matrix, Eq. (1)–(2)	Validated
ρ_{conf}, Eq. (3)	Validated
Ceiling reduction, Eq. (6)	Validated
Gap-shrinkage report	Validated
Argmax-inversion check	Validated
T^* non-monotonicity	Validated (shape finding)
$R_{\text{TS}}^{\text{conf}}$, Eq. (4)	Partially supported
Crossover horizon / κ	Partially supported
Drift sign rule (Sec. VII-E)	Negative result

the classifier-noise effect, and the residual prediction error is inherited asymptotic-bound bias rather than error introduced by this extension. Two qualifications belong in the same breath. The claim is relative to the base bound: absolute errors of 59.7% and 40.5% remain large, and the base bound is not newly accurate because of this result. The test quantifies a pre-existing limitation transferring cleanly; it does not resolve it.

The margin is 1.8%, and the pre-registered bootstrap interval for R reaches to 9.55, nearly $6.4\times$ past the threshold and well into Case 2 territory. Reporting that interval was required by the protocol, not optional context. A different draw of held-out environments could plausibly have produced Case 2. The fit-uncertainty bootstrap is also informative: the noisy-condition fit is markedly less stable than the transferred one, with a wider and lower interval.

Per-cell results (Table V) show why the medians should not be read as a tight summary. Errors are strongly heterogeneous.

Two environments over-predict by roughly $5.75\times$ under both constants and one under-predicts by roughly $3\times$. No single scalar constant corrects this spread, and consistent with the protocol's no-rescue commitment, no attempt was made to introduce one.

The fitted transfer constant, 0.0444, falls inside the pre-committed calibrated-predictor range, at its upper edge. We therefore describe the component as a calibrated predictor with theoretical motivation, not a validated bound.

D. Classification

We classify the crossover extension as *partially supported*: neither validated nor unvalidated. Case 1 is the pre-committed reported outcome and this is real held-out evidence for the first time in this line of work, a genuine improvement over the superseded in-sample analysis. The fragility is equally real. The accurate one-line summary is that the result satisfies the pre-registered criterion but not robustly, and we prefer a three-tier vocabulary precisely because a two-tier one would round this up to "validated." Table VI summarises the status of every component.

E. Drift: a documented negative result

A separate extension asks whether a low-dimensional summary statistic can predict the *sign* of adaptive advantage under environment drift, out of sample. The two extensions are independent; nothing in one affects the other's validity, and their assumption numbering is kept deliberately separate. The falsifiable structure was fixed before any attempt: a candidate statistic must beat a majority-class guess under leave-one-cell-out evaluation. Three candidates were pre-registered and tested in order, with a fourth ruled out in advance if the third failed. The substrate is a dense pre-simulated grid of 12,800 cells over four drift mechanisms.

TABLE VII
THREE PRE-REGISTERED ATTEMPTS AT A QUANTITATIVE SIGN RULE UNDER DRIFT.

#	Statistic	Outcome
1	survival, gap-change	$R^2 = 0.012$; insufficient
2	magnitude, recovery-ratio	$R^2 = 0.579$–0.735; retracted
3	environment regret, reversion	$R^2 = 0.676$; sign accuracy 79.7% = majority baseline

Results are summarised in Table VII. The second attempt was retracted before being reported as a finding, because its recovery-ratio term has the base predictor in its denominator: testing whether it predicts adaptive advantage was substantially testing whether the base bound has some validity applied per row, not independent confirmation of a new descriptor. Statistical circularity was absent; conceptual circularity was present and material.

A methodological artifact surfaced along the way and is worth stating separately. The first-attempt statistics were computed as means over touched rows only, blind to how many of an environment's rows were touched. The apparent contribution of magnitude in the second attempt ($+0.156\ R^2$) was substantially an artifact of that exclusion; recomputing at the environment level collapsed the apparent gain to $+0.006$. The environment-level quantity was verified numerically identical to an existing regret-rate function to 2.78×10^{-16} as a construction-identity check, and was never used predictively. The general lesson is that a statistic computed at the wrong level of aggregation will masquerade as needing an extra explanatory variable; the fix is recomputation, not a patch variable.

One finding does survive, and we report it as a standalone positive result separated from the failed quantitative rules: the sign of adaptive advantage under drift is governed by

TABLE VIII
ESTIMATED VERSUS ACTUAL WALL-CLOCK FOR THE FILTERED 204-CELL SWEEP.

Original estimate	38.4 min
Actual total (two attempts)	≈ 81.9 min
Ratio	$\approx 2.13\times$
Consistent with early-rate check ($\approx 2.3\times$)?	Yes

whether staleness damage *reverts*, not by its magnitude. All 13 negative-advantage cells sit at a reversion fraction of at least 0.68, spanning 260 independent seed draws; all 32 cells with reversion near zero are positive, spanning 640 draws. For the monotone-decay mechanism the reversion fraction is near zero with a 95% interval of $[0.0000, 0.0160]$ at $n = 236$, which is precise, though the sample is small and we state that as part of the finding rather than as a footnote. This qualitative pattern is real and robust even though no quantitative threshold built on top of it survives out-of-sample testing.

The pattern also explains a previously unexplained inversion: the static table beats the adaptive learner in 10 of 16 partial-reorder cells. Under that mechanism, swaps between the best and second-best action revert on a period-2 cycle, so a static table's stale action retains near-optimal value most of the time while the learner pays real exploration cost chasing a target that keeps returning to where it already was. We state this as a correlational explanation supported by a mechanism, not a causal claim. The mechanism was designed before this analysis and was never independently varied; a causal test would require drift processes matched on damage magnitude with reversion varied independently, which we did not run.

Finally, the scope statement. Drift-type fixed effects retain $+0.092$ R^2 beyond the two-statistic summary, which is real unaccounted structure. Candidate explanations we did not operationalise, having committed to stopping at three attempts, include disruption abruptness, within-row breadth, and available recovery time relative to shift frequency. Our conclusion is that the sign of adaptive advantage under drift is not reducible to a two-statistic summary across these four mechanistically distinct regimes, given the data and methods available here.

VIII. Reproducibility and Measurement Findings

The following are dated findings about how compute-cost estimation and long-run monitoring behaved in practice, independent of the theory. We include them because they invalidate cost figures we ourselves published.

A. An estimator defect

The sweep script's cost estimator calibrates single-threaded throughput on a short sample and then divides by the requested worker count, assuming linear parallel speedup; the assumption is stated in the function's own docstring. It does not hold on our hardware and was optimistic across three separate attempts. Table VIII reconciles the final run.

B. A hypothesis proposed, tested, and disproven

Our initial explanation was that no thread-limiting environment variables were set anywhere in the codebase, allowing each worker process to oversubscribe the physical cores through its own linear-algebra thread pool. We tested this directly by dispatching through the exact parallel call the sweep uses and reading the thread-pool state from inside a worker, and found it false: the backend already sets those variables to one per worker automatically, confirmed at two worker counts. The hypothesis was retracted immediately, with the disproof reported alongside it.

The underlying estimator defect was deliberately *not* retracted along with the mechanism. A four-physical-core machine under hyperthreading is sufficient explanation on its own; no claim about linear-algebra threading is required.

C. Measured effective parallelism, and its variance

We measured effective parallelism as calibrated CPU-seconds of work divided by observed wall-clock, from result-store write timestamps in one case and the script's own timer in the other. Two measurements were obtained: $1.90\times$ (60 cells, 2,843 CPU-seconds over 1,492 seconds) and $2.74\times$ (144 cells, 9,212 CPU-seconds over 3,367.45 seconds).

These disagree by about 44% on the same machine, the same workload shape, roughly 45 minutes apart. We ran no controlled comparison to attribute the gap and leave it as an open finding rather than averaging it away. Concurrent foreground work during the first window is the leading candidate but is not confirmed.

The implication for our published cost figures is direct. The headline costs of approximately 73 hours and 66 hours for the two full sweeps rest on the same falsified linear-scaling assumption. They remain un-re-derived and are suspect in both directions: the estimator over-assumes achievable speedup, which makes them optimistic, while nothing rules out worse contention at that larger scale. We flag this as follow-up work rather than correcting it here.

D. Measurement hygiene

Concurrent compute on the same machine during a timed run is an unrecorded confound that applies to every timing number this project has reported, including an earlier per-decision runtime benchmark. Whether that benchmark was collected under clean conditions is now stated as unknown rather than assumed clean: no process-isolation protocol was documented at the time, and nothing in the benchmark script records contemporaneous machine load. For future timed runs we adopted a four-point protocol: no concurrent compute during the window, no parallel tool invocations during the window, record machine load alongside timing output, and take multiple calibration draws, since two single draws under nominally identical conditions differed by $3.4\times$.

E. Non-randomised cell ordering

Grid construction orders cells by environment, then classifier, then method, then seed, without shuffling. Because the

two classifier conditions differ in throughput by roughly $218\times$, every environment's cheap cells finish before its expensive ones begin, so any cumulative-average projection taken early in a run is systematically optimistic and converges only as the observed mix approaches the true mix. Two fixes are recorded and neither is implemented: shuffle before dispatch, or project per cost-dominant population rather than pooled.

F. A stop-rule correction

An operational stop threshold intended to catch a third bad *estimate* was applied literally to a reprojection derived from a genuine *measurement*, and a run was killed. We subsequently identified this as a scope error: much larger estimates had already been accepted earlier in the same effort, so the run was never over budget, only over a number that had never been scoped to mean budget. A measurement should have been treated differently from an estimate on principle, rather than compared against the same numeric threshold. The run was relaunched immediately, resumed cleanly from banked cells, and was allowed to finish without a new threshold. We record this because it is a process correction made mid-experiment, and reporting it seems more useful than absorbing it silently.

IX. Limitations

The most restrictive limitation is scope. The held-out evaluation is confined to one matrix family at a single gap level; generalisation across other gap structures and independently generated environment families is untested. Eight held-out environments is a small test set, and the per-cell spread in Table V shows the median is not a tight summary of it.

The adaptive-side result is a point estimate that clears its threshold by 1.8% with a bootstrap interval spanning $[1.00, 9.55]$. We regard the interval as the more informative number and have tried not to let the favourable point estimate carry more weight than it can bear.

The extension inherits every limitation of the base bound: it is asymptotic rather than finite-time, carries an empirically fitted constant, and was itself reliable only in a narrow regime. Absolute held-out errors near 40 and 60 percent should be read accordingly.

Assumption A3 forbids classifier drift within the analysed horizon, which is optimistic for any deployment where the classifier degrades. Assumption A5's scalar-accuracy convenience is a real simplification, mitigated but not removed by validating against the actual confusion matrix.

For the drift work, the surviving positive finding is correlational with a mechanistic explanation, not a causal result. The mechanism whose behaviour it explains was designed before the analysis and never independently varied. Residual structure of $+0.092$ R^2 attributable to drift type remains unexplained.

Finally, the cost figures reported elsewhere in this project rest on a falsified assumption and have not been re-derived, and timing numbers collected before the hygiene protocol was adopted have unknown collection conditions.

X. Conclusion and Future Work

We extended a crossover-horizon predictor to the setting our system actually occupies, in which recovery policies observe a classifier's prediction rather than the true failure type. The extension is exact where it can be, reduces identically to its predecessor at a perfect classifier, and separates classifier-induced degradation into a ceiling channel and a learnability channel, with the former roughly six times the latter at our classifier's measured accuracy. We also showed that the predicted horizon is not monotonic in accuracy in general, which rules out an intuitive design heuristic.

The free constant was evaluated under a protocol fixed before the data existed. The reported outcome is the favourable one by point estimate, and we have tried to report it at exactly the strength the evidence supports: the pre-registered criterion is satisfied, the same protocol's uncertainty quantification does not confirm that satisfaction is robust, and the component is described as a calibrated predictor rather than a validated bound. The companion drift investigation yielded no quantitative sign rule and is reported as a negative result with one surviving qualitative finding.

Several directions follow. Extending the held-out evaluation to independently generated environment families at multiple gap levels would address the narrowest limitation. Relaxing the stationary-confusion assumption to allow classifier drift would connect the two extensions, which are currently independent. On the drift side, a causal test would require drift processes matched on damage magnitude with reversion varied independently. Finally, the project's cost figures should be re-derived from measured rather than assumed parallelism, and the earlier runtime benchmark re-collected under the hygiene protocol.

Acknowledgment

[DETAIL REQUIRED]

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